

MEScan360: A Memory-Enhanced Scanpath Prediction Model for Omnidirectional Images

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Abstract—Scanpath prediction for omnidirectional images (ODIs) aims to capture the dynamic human visual attention. However, the complicated gaze behavior and inevitable projection distortion make scanpath prediction in ODIs extremely challenging. Most existing models neither capture the long-term dependencies across visual states nor fully incorporate historical memory information, leading to limited performance. To this end, we propose MEScan360, a memory-enhanced scanpath prediction model for ODIs. We introduce two key innovations: long-term memory storage unit and memory interaction module. These two components establish a more explicit link between past visual information and current visual inputs, thereby significantly enhancing the performance of scanpath prediction. Furthermore, a robust feature extraction module is designed to extract semantic feature precisely from distorted ODIs with a more lightweight structure. Extensive experiments on several benchmark datasets demonstrate that our proposed model achieves competitive performance in both accuracy and efficiency.

Index Terms—Scanpath prediction, long-term memory storage unit, memory interaction module, omnidirectional images, robust features.

I. INTRODUCTION

Scanpath, including saccades and fixations, represents the sequential of gaze points while human exploring a visual scene. In contrast to visual saliency, scanpath characterizes the dynamic and stochastic gaze behavior as the eyes transition between points of interest in images, emphasizing both the temporal and spatial progression of visual attention throughout the observation. Studies [1] have demonstrated that human scanpaths exhibit a certain level of similarity across different individuals. Therefore, by modeling and analyzing scanpaths

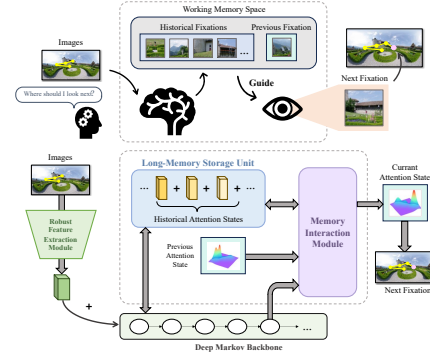


Fig. 1. We propose a memory-enhanced scanpath prediction model inspired by human visual attention mechanism.

information, researchers can gain insights into the distribution of visual attention from a different perspective.

Significant progress has been made in the exploration of scanpaths for traditional 2D images. However, with the rapid advancement of AR/VR technologies, users can now visually explore immersive 3D environments. This has led to a growing need to understand how human scanpaths move within Omnidirectional Images (ODIs), in order to optimize user experiences in virtual environments. Different from 2D images, studying human visual attention in ODIs is rather challenging. Users' head movements not only add complexity to gaze behavior but also introduce additional degrees of freedom in visual exploration. Furthermore, ODIs are usually represented by equirectangular projection, which maps the latitude and longitude of a spherical image onto horizontal and vertical grid coordinates. This method introduces substantial distortions, leading to severe stretching of objects particularly in the polar regions. This poses a challenge for existing con-

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volutional neural network-based scanpath prediction methods, as standard convolutional layers struggle to effectively process these distorted projections.

To the best of our knowledge, existing scanpath prediction methods can be generally categorized into two categories: saliency-based [2]–[4] and generative methods [5], [6]. Saliency-based methods apply a strategy that samples fixations from generated saliency map. Consequently, its performance is highly dependent on saliency maps [2], which makes it challenging to learn temporal information. How to construct an appropriate sampling strategy that accounts for time-varying features remains unclear for researchers. Generative methods typically generate scanpaths by learning human visual viewing behaviors from human data. However, these models often exhibit poor prediction stability and tend to show limited focus on points of interest during experiments. Similar to previous methods, it is also uncertain whether generative models can effectively capture and learn temporal relevance from raw data.

In dynamic visual scenes, the position of previous fixations significantly influences subsequent fixations. Therefore, the scanpath prediction task aims to generate a temporally ordered sequence of fixations, with some researchers focusing on developing methods that incorporate temporal dependencies, e.g., recurrent neural networks (RNNs) [7], [8] and long short-term memory (LSTM) [9]. Although these methods can capture the sequential dynamics in the scanpath generation, none of them captures long-range temporal dependencies across fixations. Given the temporal and spatial relationship of scanpaths, where previous visual state is crucial in determining the visual state at current moment. Moreover, research in cognitive science [10] has shown that human visual attention is guided by a complex interplay of multiple factors, such as the semantics of the observed scene and historical memory, making it an intricate task to achieve prediction results that closely align real human viewing behavior.

Inspired by neuroscience research on the role of prior history and scene semantic in guiding visual attention (see Fig. 1), we propose MEScan360, a generic scanpath prediction model that appropriately retains old memories in the visual working memory space to enhance long-range temporal dependencies in ODIs. Notably, MEScan360 is surprisingly effective, as it incorporates a long-term memory storage unit and a memory interaction module, establishing an explicit link between the current attention state and past state history. Specifically, MEScan360 includes three key designs. First, we elegantly design a long-term memory storage unit that incrementally updates the memory of latent attention state information. Second, we propose a memory interaction module that explicitly exchanges past historical visual states with the current attentional information, inherently integrating them to capture long-range temporal correlations across latent attention states. Third, we propose a robust feature extraction module to extract rotation-invariant semantic features, which can effectively improve prediction performance.

Our contributions can be summarized as follows:

- We propose a generic scanpath prediction model that

effectively preserves historical memories in the visual working memory space, which establishes long-range temporal dependencies across latent attention states and generates realistic visual viewing trajectories.

- Motivated by the role of prior historical information in guiding visual attention, as explored in neuroscience studies, we propose a long-term memory storage unit and a memory interaction module. These components establish a more explicit link between the past visual information and current visual inputs, significantly improving prediction results.

- We apply the proposed model to scanpath prediction task across four datasets. Experimental results demonstrate that our MEScan360 outperforms the state-of-the-arts methods while maintaining a more lightweight method.

II. RELATED WORKS

A. Modeling Spatial Distribution of Visual Attention

Saliency Prediction for 2D Images. Classic saliency prediction methods for 2D images primarily rely on low-level bottom-up features. There are also high-performance saliency prediction approaches that incorporate top-down features, but obstacles still exist. With the technological breakthroughs of recent years, deep learning methods [11], [12] that take advantage of large-scale databases and well-established convolutional neural networks (CNNs), have progressively emerged as the mainstream approach.

Saliency Prediction for ODIs. Existing saliency prediction methods for ODIs can be broadly classified into two main categories. viewport-based saliency prediction methods [13], [14] divide the ODI into viewports based on head movements and predict visual saliency on the local viewport plane. projection-based methods [15]–[17] relies on two widely-adopted projection methods: cubemap projection (CMP) [15] and equirectangular projection (ERP) [16], [17]. ODIs are first projected onto planes, and then specific-designed saliency detection strategies are applied for projected planes.

B. Modeling Dynamic Visual Attention

Scanpath Prediction for 2D Images. To characterize the dynamic and stochastic aspects of visual attention, researchers have introduced the scanpath prediction task. Various deep learning approaches have been proposed on 2D images. Some methods draw inspiration from biological structures [18], [19] of human eyes, extracting low-level features from the image to guide scanpath prediction. Another category is rooted in statistical principles [20], [21], establishing probability distribution of visual attention and employing techniques, e.g., component analysis or Markov chains. In contrast to interpretable models, engineering models that directly build deep learning architectures [2], [22] to fit data, have also demonstrated competitive performance.

Scanpath Prediction for ODIs. Current scanpath prediction models of ODIs can be classified into two types: saliency-based models [2]–[4] and generative models [5], [6]. The former strategies sample scanpaths from the saliency map by maximizing information gains or using a stochastic approach.

Classic network SaliNet [2] generates a temporally weighted saliency map called saliency volume. Random sampling is subsequently performed on saliency volumes to generate the scanpaths. It does take temporal information into consideration but fails to capture the equator bias feature. The latter strategies model scanpaths by learning from extensive generated scanpath data. PathGAN [6] and ScanGAN360 [5] are both built on cGAN architecture but struggle with the inherent instability of GANs. Another key challenge researchers have focused on is how to effectively incorporate the temporal dependency of fixation sequences into scanpath prediction models.

III. PROPOSED MODEL

A. Overview

Scanpaths. Scanpaths typically consist of a series of fixation points at different time steps $t \in [1, T]$. To prevent our model from misinterpreting coordinate points in spherical space, we represent fixation points on ODI using 3D Cartesian coordinates (x_t, y_t, z_t) , namely:

$$\mathbf{x}_{1:T} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T) \in \mathbb{R}^{3 \times T}. \quad (1)$$

Latent Attention States. We propose a generic scanpath prediction model that effectively preserves historical memories in the visual working memory space, which establishes long-range temporal dependencies across latent attention states. The detailed structure of MEScan360 is illustrated in Fig. 2 (a). To represent the latent attention states, we adopt a series of hidden states $\mathbf{z}_{1:T}$:

$$\mathbf{z}_{1:T} = (\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_T) \in \mathbb{R}^{n \times T}, \quad (2)$$

where n denotes the dimensionality of the attention states.

Loss Function. We employ a deep Markov network as the backbone to capture the temporal dependency of the predicted scanpath sequence. Our backbone network consists of two primary modules:

$$\text{Gated Transition} : \hat{\mathbf{z}}_t \sim p_\theta(\hat{\mathbf{z}}_t | \mathbf{z}_{t-1}), \quad (3)$$

$$\text{Emission} : \mathbf{x}_t \sim p_\theta(\mathbf{x}_t | \mathbf{z}_t). \quad (4)$$

The gated transition module obtains the raw output $\hat{\mathbf{z}}_t$ from the previous attention state $\mathbf{z}_{t-1}$. The memory interaction module is designed to get a memory-enhanced attention state $\mathbf{z}_t$ from $\hat{\mathbf{z}}_t$. The emission module converts the latent attention state $\mathbf{z}_t$ into the corresponding fixation point x_t . θ denotes all of the involved parameters, and $\sim$ is the sampling operation.

We maximize the log-likelihood function $\log(p_\theta(\mathbf{x}))$ to approximate the realistic scanpaths. The specific function is defined as follows:

$$\max \{\log p_\theta(\mathbf{x})\} = \mathbb{E}_{p_\theta(\mathbf{z}|\mathbf{x})} \frac{p_{\theta_e}(\mathbf{x}|\mathbf{z})p_{\theta_t}(\mathbf{z})}{p_\theta(\mathbf{z}|\mathbf{x})}, \quad (5)$$

The posterior $p(\mathbf{z}|\mathbf{x})$ is difficult to compute but can be approximated using variational inference. The purpose of variational inference is to derive a distribution $q_\phi(\mathbf{z}|\mathbf{x})$ that approximates

the true posterior $p_\theta(\mathbf{z}|\mathbf{x})$, s.t. $q_\phi(\mathbf{z}|\mathbf{x}) \approx p_\theta(\mathbf{z}|\mathbf{x})$. We adopt the Evidence Lower Bound(ELBO) of $\log(p_\theta(\mathbf{x}))$ as the loss function $\mathcal{L}(\mathbf{x}; \theta; \phi)$ to be maximized, namely:

$$\mathcal{L}(\mathbf{x}; \theta; \phi) = \mathbb{E}_{q_\phi(\mathbf{z}_{1:T}|\mathbf{x}_{1:T})} [\log p_{\theta_e}(\mathbf{x}_{1:T}|\mathbf{z}_{1:T})] - \mathbb{E}_{q_\phi(\mathbf{z}_{1:T}|\mathbf{x}_{1:T})} [\log q_\phi(\mathbf{z}_{1:T}) - \log p_{\theta_t}(\mathbf{z}_{1:T})], \quad (6)$$

$$q_\phi(\mathbf{z}_{1:T}|\mathbf{x}_{1:T}) = \prod_{t=1}^T q_\phi(\mathbf{z}_t | \mathbf{z}_{t-1}, \mathcal{H}(\mathbf{x}_{t:T})), \quad (7)$$

where $\mathcal{H}(\cdot)$ denotes a variational parameter computed by mapping a variable-length sequences $\mathbf{x}_{t:T}$ to m -dimensional space using RNN.

B. Robust Feature Extraction Module

In this subsection, we describe in detail the process of robust and accurate feature extraction from ODIs.

When ODIs are projected onto 2D planes, projection distortions are inevitable, particularly in the polar regions. Existing methods that rely on conventional CNNs for feature extraction often struggle to effectively address distortions. To overcome this issue, we introduce the robust feature exaction module. Specifically, our robust convolution filters work on the $p4$ group, incorporating two fundamental operations: translation and rotation. Each filter consists of all possible combinations of translations and 90° rotations around different rotation centers within a small square grid. This enables our convolution layer to effectively handle rotated features, mitigate projection distortions in ODIs, and accurately extract semantic features. To reduce the computational cost, the convolution operation is divided into two parts: the $K(\cdot)$ operation (kernel generator) and the $V(\cdot)$ operation (encoder). $\odot$ means element-wise product. Detailed definition is defined as follows:

$$f^{(l+1)}(g) = \sum_{\tilde{g} \in \mathcal{N}(g)} K_{g^{-1}\tilde{g}} \left(\bigcup_{g' \in \mathcal{N}_1(g)} f^{(l)}(g') \right) \odot V \left(\bigcup_{\tilde{g} \in \mathcal{N}_2(\tilde{g})} f^{(l)}(\tilde{g}') \right), \quad (8)$$

where $f^{(l)}$ represents the input features of l^{th} layer. Notably, feature aggregation is only performed on $\mathcal{N}(g)$, the spatial-wise neighborhood of g , i.e., $\mathcal{N}(g) = \{g(v, e_{\mathcal{A}}) \mid v \in \Omega\}$. $\Omega \in \mathbb{R}^3$ and $e_{\mathcal{A}}$ is the identity element of group $\mathcal{A}$. Similarly, $\mathcal{N}_i(g)$ ($i = 1, 2$) denotes pixel neighborhood $\{gg' \mid g' \in \mathcal{N}_i(e)\}$ and $\mathcal{N}_i(e)$ is the predefined neighborhood of the identity element e in group $\mathcal{G}$. g' and $\tilde{g}'$ represent a pair of relative locations. $\cup$ denotes the rotation-wise concatenation. Through such design, we could greatly improve computational efficiency and significantly reduce projection distortion.

C. Gated Transition

The gated transition module $p_\theta(\hat{\mathbf{z}}_t | \mathbf{z}_{t-1})$ is designed for capturing the probabilistic transition from the previous attention state $\mathbf{z}_{t-1}$ to the current state $\hat{\mathbf{z}}_t$. Neuroscience studies [10] show that semantic information plays a crucial factor in guiding human visual attention. Therefore, we incorporate

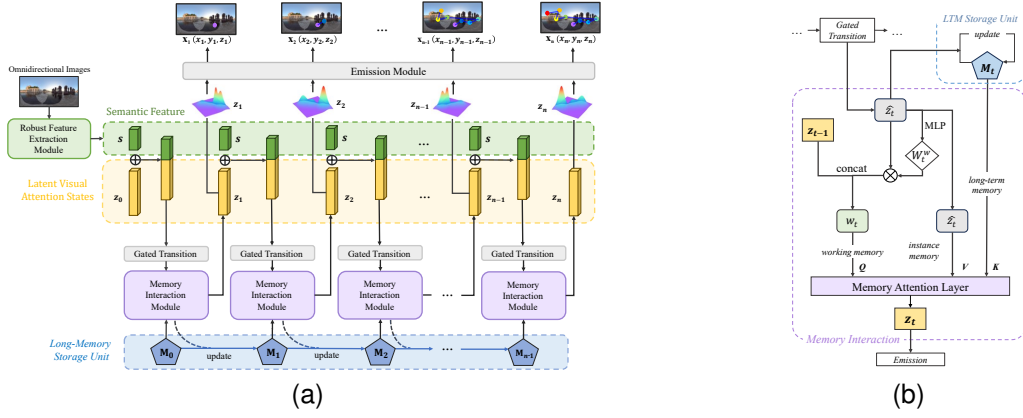


Fig. 2. (a) Overview of our proposed MEScan360. (b) Detailed structure of the memory interaction module and the long-term memory storage unit.

the influence of scene semantic feature s from robust feature extraction module into the input vector h_t :

$$h_t = W_t^h(z_{t-1} \oplus s) + b_t^h, \quad (9)$$

Then $\hat{z}_t$ is sampled from a Gaussian distribution with μ_t^z (mean) and σ_t^z (scale) as parameters. W_t and b_t are learnable parameters of the neural networks. The detailed process is described as follows:

$$\begin{aligned} \hat{z}_t &\sim \mathcal{N}(\mu_t^z, \sigma_t^z), \\ \mu_t^z &= W_t^\mu h_t + (1 - W_t^\mu)z_{t-1}, \\ \sigma_t^z &= \log(1 + e^{W_t^\sigma h_t + b_t^\sigma}). \end{aligned} \quad (10)$$

After that, the raw output $\hat{z}_t$ is sent to further processing within the memory interaction module in III-E.

D. Long-term Memory Storage Unit

In contrast to traditional methods that fewer take long-term temporal factors into consideration, we explicitly introduce a long-term memory storage unit M_t , aiming at capturing temporal correlations between latent attention states. Specifically, we initialize the long-term memory with the initial fixing state z_0 . As time progresses, shifts in visual attention states should be smooth and continuous, with minimal variations across adjacent time steps. Therefore, we assume that the semantic features of fixation states undergo only slight variations over short periods. Similarly, our designed the long-term memory we established should seamlessly update over time. A simple yet effective decaying weighted running average is used to update the long-term memory from the initial moment:

$$M_{t+1} = (1 - \lambda)z_t + \lambda M_t, \quad (11)$$

where M_{t+1} is the updated long-term memory at subsequent moment, z_t is the current visual attention state attained from the memory interaction module. To ensure smooth and consistent memory updates over time, the memory update rate is set to a small value $\lambda = 0.1$.

E. Memory Interaction Module

The memory interaction module acts as a bridge, linking current visual information with historical states. In this subsection, we will discuss on building reliable dependencies across different memory mechanisms.

Working Memory. In cognitive science and neuroscience research [10], working memory extends the concept of instance memory, emphasizing not only the short-term storage of information but also the past experiences required for its manipulation. The instance memory $\hat{z}_t$, generated by the gated transition module, is stochastic and unreliable. To improve its reliability, we introduce a learnable weight W_t^w through a simple multi-layer perceptron (MLP).

$$W_t^w = \text{Sigmoid}(\text{MLP}(\hat{z}_t)). \quad (12)$$

By multiplying W_t^w with the primitive output $\hat{z}_t$, we obtain a strengthened instance memory. To incorporate past experiences into the working memory, we introduce the previous attention state z_{t-1} , enriched with historical information:

$$\mathbf{w}_t = z_{t-1} \oplus W_t^w \hat{z}_t, \quad (13)$$

where $\mathbf{w}_t$ denotes the working memory, and $\oplus$ represents the concatenation operation.

Memory Attention Layer. We employ a multi-head attention structure as the key component of the memory interaction module, aiming to flexibly integrate the complicated memory information that has already been acquired. For the inputs to multi-head attention layer, K (key) is a reliable long-term memory M_t (as introduced in III-D), which will be continuously updated as the prediction progresses. While the outputs of the working memory and the raw output $\hat{z}_t$ are used as Q (query) and V (value), respectively.

$$\begin{aligned} Q &= \mathbf{w}_t, \quad K = \hat{z}_t, \quad V = M_t, \\ z_t &= \text{Attention}(Q, K, V). \end{aligned} \quad (14)$$

As is shown in Fig. 2 (b), the final output z_t is a more reliable attention state, incorporating an augmented memory factor that includes both working memory and long-term memory.

F. Emission Module

The emission module $p_\theta(\mathbf{x}_t|\mathbf{z}_t)$ models the transition from the latent attention state $\mathbf{z}_t$ to the fixation coordinates $\mathbf{x}_t \in \mathbb{R}^3$. After receiving the memory-augmented output from the memory interaction module, we build a Gaussian distribution determined by the reliable current attention state $\mathbf{z}_t$. Similarly, we adopt two Gaussian parameters μ_t^x (mean) and σ_t^x (scale), to describe the Gaussian distribution that x_t sampled from:

$$\begin{aligned}\mathbf{x}_t &\sim \mathcal{N}(\mu_t^x, \sigma_t^x), \\ \mu_t^x &= 2\sigma(W_x^\mu \mathbf{z}_t + b_x^\mu), \\ \sigma_t^x &= \log(1 + e^{W_x^\sigma h_t + b_x^\sigma}),\end{aligned}\quad (15)$$

where W_x and b_x denote the learnable parameters of the neural networks. Subsequently, we concatenate the outputs $\mathbf{x}_t$ at different time steps, ultimately generating the predicted scanpath $\mathbf{x}_{1:T}$ for the input ODI.

IV. EXPERIMENTS

A. Experimental Settings

Datasets. Four ODI datasets are used to validate the effectiveness of our model’s performance on scanpath predication task: Salient360! [23], Sitzmann [1], AOI [24], and JUFÉ [25]. It is worth noting that we sample the original scanpaths from Sitzmann and Salient360! at a frequency of 1 Hz to obtain the ground-truth scanpaths. For AOI and JUFÉ, we directly use the preprocessed scanpaths provided by the authors.

Evaluation Metrics. We evaluate the scanpath similarity between the ground-truth and the generated scanpaths using three metrics: Levenshtein distance (LEV), dynamic time warping (DTW), and recurrence measure (REC). Lower LEV and DTW value, higher REC value indicate better prediction accuracy. We compute the human baseline(labeled "Human") by comparing each ground-truth scanpath with the other ground-truths and averaging the results.

Implementation details. Our training phase is conducted on the Salient360! dataset. Due to the relatively small size of the training dataset, we employ a data augmentation technique to cut an ODI into six sub-images representing different perspectives. The learning rate is set to $3e-4$ and decreases by a factor of 0.99998 at each epoch. The dimensions of the visual state and RNN hidden state are set to 200 and 600, respectively. The long-time lambda is set to 0.1. Our proposed model is trained for 500 epochs.

B. Comparison with State-of-the-art Models

Model Selection. We compare our model with five state-of-the-art scanpath prediction models: CLE [20], DeepGazeIII [22], SaltiNet [2], ScanGAN [5], ScanDMM [26], and Pathformer3D [27]. We apply BMS360 [13] to generate the saliency maps for CLE. Notably, SaltiNet and DeepGazeIII are built on 2D image datasets and are subsequently fine-tuned on the ODI dataset. The latter three models are specifically designed for scanpaths prediction tasks on ODIs.

Quantitative Comparison. Table I shows the results of quantitative comparison, with the best results highlighted in

TABLE I
THE RESULT OF QUALITATIVE COMPARISON, THE BEST PERFORMANCE IN EACH METRIC IS HIGHLIGHT IN BOLD.

Dataset	Method	LEV↓	DTW↓	REC↑
Sitzmann [1]	CLE	44.683	1977.250	1.891
	DeepgazeIII	46.424	1992.859	3.082
	SaltiNet	51.370	2305.099	2.364
	ScanGAN	45.270	1951.848	3.241
	ScanDMM	44.966	1965.427	3.475
	Pathformer3D	45.825	1932.249	3.609
	MEScan360 _M	45.062	1906.803	4.014
	MEScan360 _L	45.434	1997.839	3.470
	Human	41.188	1836.986	6.345
Salient360! [23]	CLE	44.683	1977.250	1.891
	DeepgazeIII	42.044	1887.664	1.899
	SaltiNet	43.076	2027.178	1.485
	ScanGAN	38.893	1703.458	3.174
	ScanDMM	38.617	1704.803	3.214
	Pathformer3D	37.762	1745.981	3.610
	MEScan360 _M	35.773	1516.423	4.540
	MEScan360 _L	37.377	1721.429	3.573
	Human	35.084	1382.590	5.202
AOI [24]	CLE	13.707	496.093	3.947
	DeepgazeIII	13.647	642.996	1.956
	SaltiNet	14.811	697.589	1.551
	ScanGAN	12.889	552.446	3.750
	ScanDMM	12.883	587.495	3.481
	Pathformer3D	16.022	642.279	3.613
	MEScan360 _M	12.635	571.875	4.014
	MEScan360 _L	12.925	591.353	3.307
	Human	9.243	389.477	6.228
JUFÉ [25]	CLE	26.275	1106.220	1.966
	DeepgazeIII	24.998	1178.953	2.149
	SaltiNet	26.074	1287.144	1.540
	ScanGAN	24.262	1102.608	3.327
	ScanDMM	23.684	1129.926	3.206
	Pathformer3D	23.338	1091.159	3.575
	MEScan360 _M	24.146	1090.217	3.720
	MEScan360 _L	24.135	1142.025	3.130
	Human	18.306	1038.880	7.745

gray. We evaluate two versions of our proposed model: the lightweight version MEScan360_L with our robust feature extraction module, and a more accurate version MEScan360_M that incorporates SCNN. MEScan360_M shows significant improvement over existing models, achieving strong performance across all metrics on the Sitzmann and Salient360! datasets. Meanwhile, MEScan360_L achieves lightweight efficiency while maintaining quantitative results comparable to state-of-the-art models.

Qualitative Comparison. To demonstrate our model’s outstanding performance and wide applicability, we selected ODIs from four datasets, each representing a different visual scene. As shown in Fig. 3, our model generates stable predictions, effectively captures salient regions and demonstrates superior performance in handling equator bias.

Efficiency Comparison. Model complexity is also a crucial metric for performance assessment. We compare the model’s efficiency in terms of model scale and average runtime for

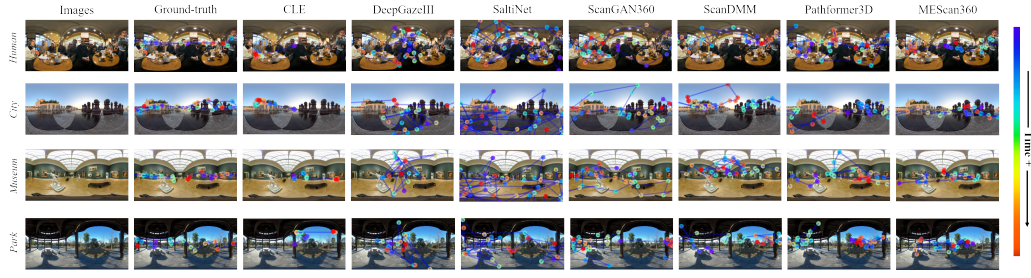


Fig. 3. The result of qualitative comparison, scanpath length are set to 20.

generating (25×30) scanpaths of an ODI. The results are shown in Table II. It is evident that our model has the smallest size (5.49 MB), and the least runtime (1.081 s).

TABLE II
EFFICIENCY COMPARISON IN TERMS OF THE MODEL SIZE AND RUNTIME.

Model	Model Size	Runtime
CLE	-	29.662s
DeepGazeIII	78.9MB	4.237s
SaltiNet	103.6MB	≈ 15 min
ScanGAN	33.9MB	10.364s
ScanDMM	17.8MB	1.807s
Pathformer3D	19.4MB	3.891s
MEScan360	5.49MB	1.018s

V. CONCLUSION

The primary goal of this paper is to propose a scanpath prediction model that retains historical information as well as establishes long-term dependency between chronological visual-attention states. We innovatively introduce long-term memory storage unit and memory interaction module, which integrate historical information into the visual working memory space and emphasize the influence of previous attention states on scanpath generation. Additionally, we incorporate a robust feature extraction module to obtain precise features from distorted ODIs. This module also significantly reduces the model's parameter scale. Extensive experiments are conducted across datasets, demonstrating our model's remarkable performance in the scanpath prediction task.

REFERENCES

- [1] V. Sitzmann, A. Serrano, A. Pavel, M. Agrawala, et al., "Saliency in vr: How do people explore virtual environments?," *IEEE transactions on visualization and computer graphics*, vol. 24, no. 4, pp. 1633–1642, 2018.
- [2] M. Assens Reina, X. Giro-i Nieto, K. McGuinness, N. E. O'Connor and N. E. O'Connor, "Saltinet: Scan-path prediction on 360 degree images using saliency volumes," in *ICCV*, 2017, pp. 2331–2338, IEEE.
- [3] M. Assens, X. Giro-i Nieto, K. McGuinness, N. E. O'Connor and N. E. O'Connor, "Scanpath and saliency prediction on 360 degree images," *Signal Processing: Image Communication*, vol. 69, pp. 8–14, 2018.
- [4] Y. Zhu, G. Zhai, X. Min, J. Zhou and J. Zhou, "The prediction of saliency map for head and eye movements in 360 degree images," *IEEE Transactions on Multimedia*, vol. 22, no. 9, pp. 2331–2344, 2019.
- [5] D. Martin, A. Serrano, A. W. Bergman, G. Wetzstein and B. Masia, "Scangan360: A generative model of realistic scanpaths for 360 images," *IEEE Transactions on Visualization and Computer Graphics*, vol. 28, no. 5, pp. 2003–2013, 2022.
- [6] M. Assens, X. Giro-i Nieto, K. McGuinness, N. E. O'Connor and N. E. O'Connor, "Pathgan: Visual scanpath prediction with generative adversarial networks," in *ECCV*, 2018, pp. 0–0.
- [7] T. Ngo and B. Manjunath, "Saccade gaze prediction using a recurrent neural network," in *ICIP*, IEEE, 2017, pp. 3435–3439.
- [8] C. Li, W. Zhang, Y. Liu, Y. Wang and Y. Wang, "Very long term field of view prediction for 360-degree video streaming," in *MIPR*, IEEE, 2019, pp. 297–302.
- [9] Z. Chen and W. Sun, "Scanpath prediction for visual attention using ior-roi lstm," in *IJCAI*, 2018, pp. 642–648.
- [10] J. M. Wolfe, "Guided search 6.0: An updated model of visual search," *Psychonomic bulletin & review*, vol. 28, no. 4, pp. 1060–1092, 2021.
- [11] G. Li and Y. Yu, "Deep contrast learning for salient object detection," in *CVPR*, 2016, pp. 478–487.
- [12] Y. Wang, W. Zhang, L. Wang, T. Liu and H. Lu, "Multi-source uncertainty mining for deep unsupervised saliency detection," in *CVPR*, 2022, pp. 11727–11736.
- [13] P. Lebreton and A. Raake, "Gbv360, bms360, prosal: Extending existing saliency prediction models from 2d to omnidirectional images," *Signal Processing: Image Communication*, vol. 69, pp. 69–78, 2018.
- [14] F. Battisti, S. Baldoni, M. Brizzi, M. Carli and M. Carli, "A feature-based approach for saliency estimation of omni-directional images," *Signal Processing: Image Communication*, vol. 69, pp. 53–59, 2018.
- [15] R. Monroy, S. Lutz, T. Chalasani, A. Smolic and A. Smolic, "Sal-net360: Saliency maps for omni-directional images with cnn," *Signal Processing: Image Communication*, vol. 69, pp. 26–34, 2018.
- [16] H. Lv, Q. Yang, C. Li, W. Dai, et al., "Salgen: Saliency prediction for 360-degree images based on spherical graph convolutional networks," in *ACM MM*, 2020, pp. 682–690.
- [17] D. Chen, C. Qing, X. Xu, H. Zhu and H. Zhu, "Salbinet360: Saliency prediction on 360° images with local-global bifurcated deep network," in *IEEE VR*, 2020, pp. 92–100.
- [18] H. Adeli, F. Vitu and G. J. Zelinsky, "A model of the superior colliculus predicts fixation locations during scene viewing and visual search," *Journal of Neuroscience*, vol. 37, no. 6, pp. 1453–1467, 2017.
- [19] D. Zanca, S. Melacci and M. Gori, "Gravitational laws of focus of attention," *IEEE transactions on pattern analysis and machine intelligence*, vol. 42, no. 12, pp. 2983–2995, 2019.
- [20] G. Boccignone, V. Cuculo and A. D'Amelio, "How to look next? a data-driven approach for scanpath prediction," in *Formal Methods. FM 2019 International Workshops: Porto, Portugal, October 7–11, 2019, Revised Selected Papers, Part I 3*, Springer, 2020, pp. 131–145.
- [21] A. Coutrot, J. H. Hsiao and A. B. Chan, "Scanpath modeling and classification with hidden markov models," *Behavior research methods*, vol. 50, no. 1, pp. 362–379, 2018.
- [22] M. Kümmerer, M. Bethge and T. S. Wallis, "Deepgaze iii: Modeling free-viewing human scanpaths with deep learning," *Journal of Vision*, vol. 22, no. 5, pp. 7–7, 2022.
- [23] Y. Rai, J. Gutiérrez and P. Le Callet, "A dataset of head and eye movements for 360 degree images," in *MMSys*, 2017, pp. 205–210.
- [24] M. Xu, L. Yang, X. Tao, Y. Duan and Z. Wang, "Saliency prediction on omnidirectional image with generative adversarial imitation learning," *IEEE Transactions on Image Processing*, vol. 30, pp. 2087–2102, 2021.
- [25] Y. Fang, L. Huang, J. Yan, X. Liu and Y. Liu, "Perceptual quality assessment of omnidirectional images," in *AAAI*, 2022, vol. 36, pp. 580–588.
- [26] X. Sui, Y. Fang, H. Zhu, S. Wang and Z. Wang, "Scandmm: A deep markov model of scanpath prediction for 360deg images," in *CVPR*, 2023, pp. 6989–6999.
- [27] R. Quan, Y. Lai, M. Qiu, D. Liang and D. Liang, "Pathformer3d: A 3d scanpath transformer for 360° images," *arXiv preprint arXiv:2407.10563*, 2024.